

3 Yr. Degree/4 Yr. Honours 4th Semester Examination, 2026 (CCFUP)

Subject : Chemistry

Course: CHEM4012 (Major)

Time: 3 Hours

Full Marks: 60

*The figures in the right hand margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

1. Answer any ten questions from the following:

2×10=20

~~(a)~~ Comment— 'Partial molal enthalpy is an extensive property and a vector quantity'.

(b) Show that chemical potential increases with rise in pressure at a constant temperature.

~~(c)~~ Ionic conductance of H^+ and OH^- is very high. Why?~~(d)~~ How can you determine Λ° of a weak electrolyte graphically? 20~~(e)~~ Find a relation between K_a and K_c for the reaction at equilibrium(f) $\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$ is an eigen function of the operator $\frac{-\hbar^2}{2m} \frac{d^2}{dx^2}$. Find the eigenvalue.

$$\left(\hbar = \frac{h}{2\pi} \right)$$

~~(g)~~ Define turn over number. Comment on its unit. ?~~(h)~~ Transport number of Cd^{2+} ion in a concentrated solution of CdI_2 may become zero or even negative. Why?~~(i)~~ Define molar conductance. How is it related with equivalent conductance?~~(j)~~ Show a plot of C_p/T vs T on consideration of the Third law of Thermodynamics. What is the meaning of the area under the curve? *Internal value*~~(k)~~ What do you mean by a 'generalised acid-base catalysis'? Give an example.

(l) What do you mean by 'complimentary variables' in quantum mechanics? Give example. ?

~~(m)~~ Prove that, $\hat{A} = i \frac{d}{dx}$ is hermitian ($i = \sqrt{-1}$).

- (n) How can $\Delta\mu$ of a reaction be determined from equilibrium study?
- (o) State the effect of introduction of an inert gas to a reaction at equilibrium at constant volume. Justify your answer.

2. Answer any four questions from the following:

5×4=20

- (a) Derive an expression of ΔS_{mix} when two or more ideal gases are mixed up at constant T and P. Find its maximum value. 3+2
- (b) At 25°C, 1 mole of acetic acid and 1 mole of ethanol are mixed. At equilibrium, 0.667 mole of acid have reacted. Calculate K_c . Also calculate ΔG_c° for the reaction. 3+2
- (c) (i) Show that an increase in the velocity by dv causes de Broglie wavelength to decrease by $\frac{\lambda}{v} dv$. Hence, show that $\frac{\lambda}{v} + \frac{dv}{v} = 0$.
- (ii) Normalize the wavefunction $\psi = A \cdot e^{ik\phi}$ defined in $(0, 2\pi)$ where A and k are constants ($i = \sqrt{-1}$). 1½×2+2
- (d) In a moving boundary experiment with 0.02N NaCl solution, a current of 0.0016 Ampere moved the boundary through a distance of 6 cm for the cation in 2070 sec. Cross-section of the tube is 0.1115cm². Calculate the transport number for Na⁺ and also of Cl⁻ ion. 5
- (e) Show that the observed rate constant for a reaction catalysed by both acids and bases with respective catalytic rate constant of K_{H^+} and K_{OH^-} will pass through a minima when H⁺ concentration is varied. Calculate the pH at which this minima is observed at 298K if $\frac{K_{H^+}}{K_{OH^-}} = 100$. Given ionic product of water at 298K is 1.0×10^{-14} . 3+2
- (f) (i) Explain whether the following functions are acceptable as quantum mechanical wave functions within the given range of variables:
- (a) $\psi_x = e^{\pm |x^3|}$, $-100 \leq x \leq 100$,
- (b) $\psi_y = \frac{y^3 - 2y^2 + 4y - 3}{6y^2 - 5y - 6}$, $-5 \leq y \leq 5$.
- (ii) Calculate the de Broglie wavelength of an electron moving with a speed of one-tenth of velocity of light. 1½×2+2

3. Answer any two questions from the following:

10×2=20

- (a) (i) Define catalytic coefficient. Find its dimension.
- (ii) Show an energy profile diagram for a catalysed and uncatalysed reaction. Comment on if there is any change in ΔH of the reaction.

- (iii) Demonstrate graphically the primary kinetic salt effect on the following reaction:



Mention the slope value.

- (iv) Prove that for an enzyme catalysed reaction, the Michaelis constant is the substrate concentration at which the rate is one-half of the maximum rate. 2+2+2+4

- (b) (i) Show the plot of conductometric titration curve when a mixture of HCl and CH_3COOH is titrated with NaOH solution. Indicate the endpoints on the plot.
- (ii) State and explain the Debye-Falkenhagen effect.
- (iii) Does equilibrium constant of a reaction depend on the representation of the reaction? Elaborate.
- (iv) How absolute value of entropy of a gas can be calculated on the basis of the Third law of thermodynamics? 2+3+2+3

- (c) (i) What is threshold frequency in photo-electric effect? State the effect of increase in intensity of light when the frequency is below and above the threshold frequency. How Einstein's equation of photo-electric effect can be used to determine the Planck's constant? 12

- (ii) State the uncertainty principle in terms of angular momentum, explaining the terms involved.

- (iii) Find the commutation $\left[\frac{\hat{1}}{x}, \hat{p}_x \right]$.

- (iv) Calculate the minimum uncertainty in velocity of an electron when the uncertainty in position is $5\text{\AA}$. 4+2+2+2

- (d) (i) Derive an expression of rate constant by using the Collision Theory of bimolecular gaseous reactions.
- (ii) Define entropy of activation. State its significance.
- (iii) State Debye's T^3 law. How far was it successful in explaining heat capacity of solids?
- (iv) Show that for an eigen system, the expectation value is the eigenvalue of the system. 3+2+(1+2)+2